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Decision Trees & Random Forests

Understanding tree-based models — from single decision trees to powerful ensemble methods that combat overfitting and improve predictive accuracy.

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CHAPTER 8.1

What Is a Decision Tree?

A **Decision Tree** is a non-parametric supervised learning algorithm used for both **classification** and **regression** tasks. Unlike linear models that fit a single equation to the entire dataset, decision trees work by constructing a tree-like structure of rules that recursively partition the feature space into increasingly homogeneous regions.

At each internal node, the algorithm selects the feature and threshold that provides the **best separation** of classes (or reduction in variance for regression). This greedy, top-down splitting continues until a stopping criterion is met — such as maximum depth, minimum samples per leaf, or perfect purity. The result is an intuitive, interpretable model that mirrors human decision-making logic.

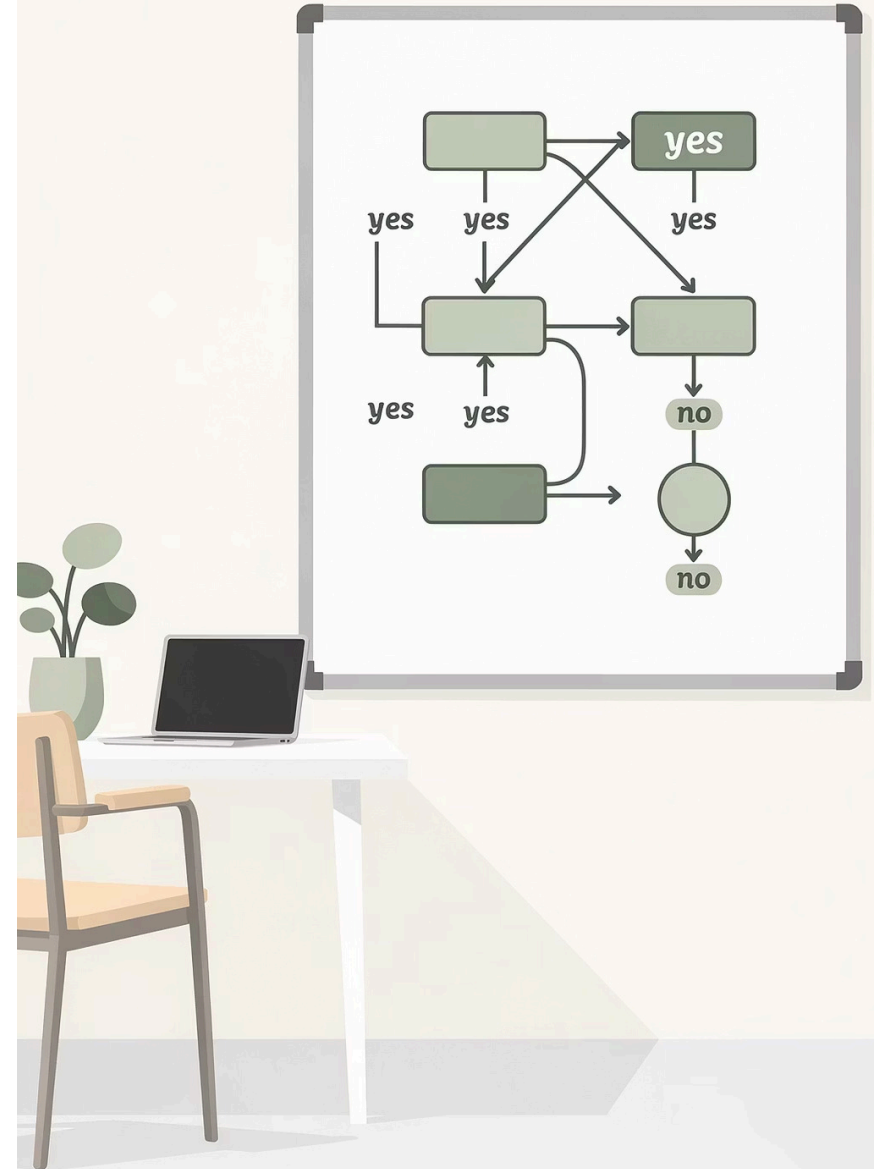
How Splitting Works

The Greedy Algorithm

At every node, the tree evaluates all possible features and all possible split points to find the one that maximizes the separation of target classes. This is a **greedy** approach — it optimizes locally at each step without considering future splits.

The quality of each candidate split is measured by an **impurity metric**. The split that produces the largest decrease in impurity is chosen. Two dominant metrics are used in practice: **Gini Impurity** and **Entropy**.

📌 **Key Insight:** Because the algorithm is greedy, a decision tree does not guarantee a globally optimal structure. Small changes in data can lead to very different trees — a property known as **high variance**.



Impurity Metrics: Gini vs. Entropy

The choice of impurity metric determines how the tree evaluates candidate splits. Both metrics reach their minimum (zero) when a node is **perfectly pure** — all samples belong to a single class.

Gini Impurity

Measures how often a randomly chosen element from a set would be **incorrectly labeled** if it were assigned a class based on the distribution of labels in that set.

Range: 0 (pure) to 0.5 (maximum impurity for binary classification)

Computationally efficient and the default in most implementations like scikit-learn.

Entropy

Quantifies the **randomness or uncertainty** in a set of data, borrowed from information theory. An entropy of 0 means the data is perfectly homogeneous; higher values indicate more disorder.

Range: 0 (pure) to 1.0 (maximum uncertainty for binary classification)

Tends to produce slightly more balanced trees in practice.

Gini & Entropy — The Math

Gini Impurity Formula

$$G(t) = 1 - \sum_{k=1}^K p_k^2$$

Where p_k is the proportion of class k samples at node t . For a binary split with 50/50 class distribution, Gini = 0.5 (maximum impurity).

Entropy Formula

$$H(t) = - \sum_{k=1}^K p_k \log_2(p_k)$$

Where p_k is the proportion of class k at node t . For a 50/50 split, Entropy = 1.0. The **Information Gain** is the parent entropy minus the weighted child entropy.

Decision Tree — Strengths & Limitations

✓ Highly Interpretable

Decision trees produce white-box models that can be visualized and understood by non-technical stakeholders. Each path from root to leaf represents an explicit set of rules.

✓ No Feature Scaling Needed

Because splits are based on thresholds, decision trees are invariant to monotonic transformations of features. No normalization or standardization required.

⚠ Prone to Overfitting

Without pruning or depth limits, a tree will grow until every leaf is pure — memorizing noise in the training data and failing to generalize to unseen examples.

⚠ High Variance

Small perturbations in training data can produce dramatically different tree structures, making single trees unstable predictors.



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CHAPTER 8.2

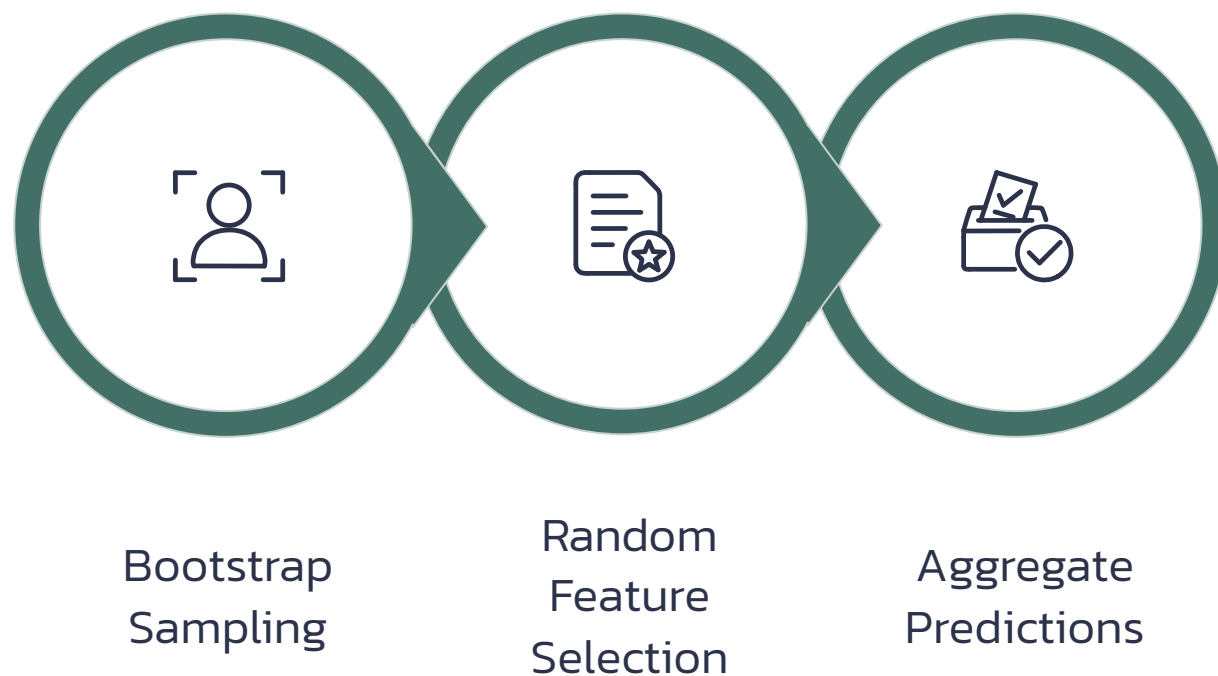
Enter Random Forests

A single decision tree's greatest weakness — **overfitting** — becomes the motivation for one of machine learning's most powerful ensemble methods. **Random Forests** address this by constructing a large collection of decorrelated decision trees and aggregating their predictions.

The core idea is elegantly simple: if individual trees are noisy but roughly unbiased, then averaging many of them together will **cancel out the noise** while preserving the signal. This principle, known as the "wisdom of crowds," reduces variance without significantly increasing bias.

How Random Forests Work

Random Forests introduce **two layers of randomness** to ensure that the individual trees in the ensemble are as different from each other as possible, maximizing the benefit of aggregation.



By training each tree on a different bootstrap sample and restricting feature choices at every split, the forest produces diverse, decorrelated trees whose combined prediction is far more robust than any single tree.

Why Ensembles Reduce Overfitting



Bagging (Bootstrap Aggregating)

Each tree sees a different random sample of the data, so individual overfitting patterns differ and cancel out when averaged.



Feature Randomness

By limiting features at each split (typically $\sqrt{p}$ for classification), trees become decorrelated — the key to effective variance reduction.



Bias-Variance Tradeoff

Individual trees have low bias but high variance. Averaging many trees preserves the low bias while dramatically shrinking variance.



Single Tree vs. Random Forest

Understanding the practical differences helps you choose the right tool for the job. Here is a side-by-side comparison across critical dimensions.

Dimension	Single Decision Tree	Random Forest
Variance	High — sensitive to data changes	Low — averaging smooths noise
Bias	Low (deep trees)	Slightly higher, but well compensated
Interpretability	Excellent — easy to visualize	Lower — hundreds of trees
Training Speed	Fast	Slower (many trees)
Generalization	Poor without pruning	Strong out-of-the-box
Hyperparameters	max_depth, min_samples	n_estimators, max_features, plus tree params

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KEY TAKEAWAYS

Key Takeaways

Decision Trees

Intuitive, interpretable models that recursively split data using impurity metrics like Gini and Entropy — but prone to overfitting without constraints.

Overfitting Problem

A single deep tree memorizes training noise. High variance means small data changes produce wildly different models.

Random Forests

Ensemble of decorrelated trees using bagging + feature randomness. Averaging reduces variance while maintaining low bias — a robust, go-to algorithm.

"Random Forests are one of the most successful general-purpose algorithms in modern machine learning — strong performance with minimal tuning."

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